JUYEON LEE

Post-Master's Researcher at Korea Brain Research Institute (KBRI)

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Education

M.S. in Brain Science, *Daegu Gyeongbuk Institute of Science and Technology (DGIST)* 2022 – 2024
Advisor: Gunsoo Kim, Ph.D. Daegu, South Korea
Graduate research conducted at KBRI

B.S. in Software Engineering, *Hanyang University* 2017 – 2022
cum laude Gyeonggi, South Korea

Research Experience

Post-Master's Researcher, *Auditory Processing Lab (KBRI)* 09/2024 – Present
Advisor: Gunsoo Kim, Ph.D. Daegu, South Korea

- **Neural Coding of Vocalizations in Background Noise:** Investigated how different background noise types affect neural encoding of vocalizations in the mouse inferior colliculus using single-unit recordings and neural response analyses.
- **Neuropixels Recording and Spike-Sorting Pipeline:** Recorded neural activity in awake mice using Neuropixels probes and established a Kilosort-based workflow for spike sorting and manual curation.
- **Auditory Behavioral Experiments:** Designed and implemented a MonkeyLogic-based Go/No-Go tone-discrimination task, built the behavioral apparatus, and assessed discrimination performance in mice during locomotion and rest.
- **Retrograde Viral Tracing and 3D Brain Reconstruction:** Performed retrograde viral injections into the mouse inferior colliculus and visualized labeled neurons using 3D brain reconstruction.
- **Mouse Pose Estimation Using DeepLabCut:** Established a DeepLabCut-based pose-estimation pipeline to track swimming mice in a collaborative project led by Dr. Youngrae Ji (KBRI).
- **Audiogenic Seizure EEG Recording:** Implanted skull screw electrodes and recorded EEG in *Strc* knockout mice.
- **Cerebellar Immunohistochemistry:** Performed dual-label immunofluorescence staining for NALCN and calbindin-D28k to examine NALCN localization in cerebellar Purkinje cells.

Research Intern, *Cognitive Behavioral Neuroscience Lab (KBRI)* 03/2022 – 08/2022
Advisor: Satoshi Kojima, Ph.D. Daegu, South Korea

- **Songbird Behavioral Analysis Using DeepLabCut:** Applied DeepLabCut-based markerless pose estimation to track songbird movements and quantify behavioral patterns.

Research Intern, *Auditory Processing Lab (KBRI)* 07/2021 – 02/2022
Advisor: Gunsoo Kim, Ph.D. Daegu, South Korea

- **MATLAB-Based Auditory Stimulus Generator:** Improved the user interface of an existing MATLAB application to enhance usability and streamline auditory stimulus generation.
- **Auditory Stimulus Reconstruction in Noise:** Designed dynamic random chord (DRC) stimuli embedded in white noise and applied stimulus reconstruction analysis to previously recorded neural data.

Researcher, *Data Science Lab (KT Affiliated Research Institute)* 09/2020 – 12/2020
Seoul, South Korea

- **Applied Data Science and Machine Learning Projects:** Applied NLP, LDA topic modeling, sentiment analysis, association rule mining (Apriori), geospatial data analysis, and collaborative filtering to consumer analytics, population analysis, and recommendation systems.

Undergraduate Research Student, *AI Design Lab (Hanyang University)* 01/2020 – 02/2020
Advisor: Dayoung Joo, Ph.D. Gyeonggi, South Korea

- **Hyundai Motor Company – EEG-Based Human–Vehicle Interaction Research:** Assisted with EEG recordings during simulated-driving experiments involving visual, auditory, and tactile stimuli and driving-related decision-making to investigate user experience in future vehicles.

Publications

Lee, J., Jisoo Han, Gunsoo Kim. Neural sound processing in noise in the mouse auditory midbrain. *(in preparation)*

dos Santos, E. B., Mori, C., Kim, Y., Kim, J., Tachibana, R. O., Jeong, O.-Y., **Lee, J**., & Kojima, S. (2026). A rebound from pre-dawn singing suppression drives intense morning singing in captive zebra finches. *Communications Biology (submitted)*.

Research Presentations & Talks

Juyeon Lee, Jisoo Han, Gunsoo Kim. Neural sound processing in noise in the mouse auditory midbrain. *KBRI Research Day* ***(Awarded Excellent Poster Prize).*** 12/2024
Daegu, South Korea

Juyeon Lee, Jisoo Han, Gunsoo Kim. Neural sound processing in noise in the mouse auditory midbrain: Context-Dependent Signal Extraction in IC Neurons. *The Korean Society for Brain and Neural Sciences (KSBNS).* 10/2024
Gyeongju, South Korea

Juyeon Lee, Jisoo Han, Gunsoo Kim. Neural sound processing in noise in the mouse auditory midbrain: Dynamic Noise Sensitivity Across Acoustic Contexts. *International Brain Initiative* (*IBI) Conference* ***(Awarded Excellent Poster Prize & Young Investigators' Talk).*** 07/2024
Daegu, South Korea

Juyeon Lee, Jisoo Han, Gunsoo Kim. Neural sound processing in noise in the mouse auditory midbrain: Neural Reconstruction of Sounds in Noise. *The Korean Society for Brain and Neural Sciences (KSBNS).* 09/2023
Busan, South Korea

Juyeon Lee, Jisoo Han, Gunsoo Kim. Neural sound processing in noise in the mouse auditory midbrain: Noise-Resistant Encoding of Natural Sounds. *International Brain Initiative* (*IBI) Conference* ***(Awarded Excellent Poster Prize)****.* 08/2023
Daegu, South Korea

Fellowships & Awards

Junior Researcher Fellowship (2025-2028), *KBRI* 05/2025
Neural Mechanisms of Auditory Perception in Noise

Best Mentee, *Women in Science and Engineering, Technology (WISET) Mentoring* 12/2020

SW Academic Conference, *Software Business Group, Hanyang University* 12/2020
Excellence Award

Industry Cluster-PBL Competition, *Hanyang University IC-PBL Center* 12/2019
Excellence Award

SW Startup IDEA Competition, *Software Business Group, Hanyang University* 09/2019
Excellence Award

Technical Skills

Electrophysiology & Surgery
In vivo extracellular recordings (tungsten microelectrodes, Neuropixels probes), stereotaxic surgery, craniotomy, viral injections, EEG recording.

Programming & Data Analysis
MATLAB, Python

Histology & Imaging
Immunohistochemistry (IHC), confocal microscopy, ImageJ/Fiji, 3D brain reconstruction (AMaSiNe).

Spike Sorting & Curation
Kilosort, SpikeInterface, Phy, Plexon Offline Sorter (OFS).

Behavioral Neuroscience
MonkeyLogic, DeepLabCut.

Molecular Biology
Polymerase chain reaction (PCR).

Teaching Experience

Teaching Assistant, *DGIST, Course: General Biology Laboratory (BS115)* 08/2024 – 12/2024

Leadership & Service

World Vision 2023 – 2025
Letter Translation Volunteer

DGIST Buddy Program 02/2024 – 06/2024
Peer Mentor for International Students

Korean-American Women in Science and Engineering (KWiSE), 2022
Mentee Leader, KWiSE Mentoring

'Low Code, No Code' Workshop 08/2020
Microsoft Devs Speak (APAC)
Presented Microsoft Power Platform and its applications in low-code/no-code app development and workflow automation.

Women in Science, Engineering and Technology (WISET), 2020
Mentee Leader, KT Mentoring Program

Women's Engineering Research Workshop, 2020 – 2022
Women in Science, Engineering and Technology (WISET)
Delivered annual presentations introducing the software engineering major and related career opportunities.